

# The Perception of Technological Risks: A Literature Review

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## ABSTRACT

In response to rising concern about technological risks, a concerted effort is being made to improve risk analysis methods and risk management approaches. As part of this effort, behavioral and social scientists have produced a substantial body of knowledge of value to risk analysts and decision makers. This paper focuses on behavioral and social science studies of human intellectual limitations in thinking about risks, factors influencing risk attitudes and perceptions, and factors contributing to social conflicts and disputes about technological activities. A basic assumption of the paper is that analysts and decision makers can benefit from a better understanding of how experts and nonexperts think and make decisions about technological risks. Without such understanding, well-intended policies may be ineffective or even counterproductive.

## Introduction

*A truly unexpected result came out of the Kemeny Commission's study of Three Mile Island. A group that set out to investigate a technology ended up talking about people. . . . In the commission's own words, "It became clear that the fundamental problems were people-related problems."*

Editorial, *Washington Post*, October 31, 1979.

Behavioral and social scientists in several countries are currently grappling with several people-related questions related to risk: What factors influence individual perceptions of risk? What accounts for anomalies in the way individuals and groups behave when faced with ostensibly comparable risks—such as the risks of nuclear power, dam failures, or earthquakes? What weight should decision makers attach to public perceptions of risk in determining how safe is safe enough? Are there ways to increase our capacity for dealing with technological risks in a rational manner?

What follows is a review of the behavioral and social science literature pertaining to these questions (See [1] through [166]). Before beginning, however, several points need to be made concerning the quality of the data and the research. First, most of the reported findings are based on surveys of small, highly specialized, and unrepresentative groups. An important set of risk perception studies undertaken by Paul Slovic and his colleagues,

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for example, relied almost entirely on data gathered from residents of Eugene, Oregon, a small university town located in a state with a high level of environmental concern and a progressive environmental protection program [29, 127, 129]. The respondents included 40 members of the Eugene League of Women Voters, 40 college students at the University of Oregon, 25 Eugene business people, and 15 persons selected nationwide for their expertise in risk analysis.

Second, little attempt has been made by researchers to analyze the effects of organizational and social structural variables (e.g., ethnicity, religion, sex, region of the country, age group, occupation, education, income, marital status, organizational membership, and organizational location) on risk perceptions. Most studies adopt a personal or technical perspective [71] and start from the assumption that individual risk perceptions can be explained by the psychological makeup of the individual or by the degree to which the individual has access to, and correctly interprets, technical information. With relatively few exceptions, researchers have not adopted an organizational or social structural perspective, which assumes that risk perceptions are substantially influenced by group norms and expectations, and by the social and organizational location of the individual. As Linstone [71] has shown, studies that ignore or unduly emphasize one of these three perspectives—personal, technical, or organizational—are considerably less useful than studies that attempt an integrated analysis. Unfortunately, risk perception research is still at an early stage of development, and this integration has not yet occurred.

Third, the risk perception literature suffers, in an exaggerated form, from shortcomings common to nearly all survey research [28]: (1) people typically respond to survey questions with the first thing that comes to mind, and then become committed to their answer; (2) people typically provide an answer to any question that is posed, even when they have no opinion, when they do not understand the question, or when they hold inconsistent beliefs; (3) survey responses can be influenced by the order in which the questions are posed, by whether the emphasis is on speed or accuracy, by whether the question is closed or open, by whether respondents are asked for a verbal or numerical answer, by interviewer prompting, and by how the question is posed. Risk perception surveys are especially vulnerable to these types of biases, because people are often unfamiliar with the activity or substance being assessed and because they may not understand the technical and methodological issues under debate.

Fourth, although risk perceptions may be inconsistent with behavior, relatively few studies have examined the relationship between perceptions of technological hazards and the behavior of people in actual situations. Empirical studies from other social and behavioral fields suggest that the linkages between perception and behavior are highly complex and appear to be mediated by several factors [6, 13, 62]. Researchers have shown, for example, that activist behavior is related to a willingness to participate in group activities, a positive identification with potential group leaders, a belief in the efficacy of social action, and physical proximity to arenas of social conflict [13, 30]. With few exceptions [78, 90], risk perception researchers have not examined these variables.

Fifth, for reasons that are not entirely clear, researchers have made few attempts to relate the literature on the perceived risks of technological hazards to the extensive literature on the perceived risks of natural disasters [2, 3, 4, 9, 10, 12, 17, 34, 51, 52, 39, 64, 71, 81, 82, 83, 86]. To date, only limited efforts have been made to replicate and extend natural hazard studies concerned with the various factors that affect perceived risks, including the perceived cause of the disaster, the degree to which risk information is available and accessible, the form in which risk information is presented, the institutional

and social location of the individual evaluating the risk, the individual's previous disaster experiences, and the perceived level of hazard protection and security. Furthermore, only limited efforts have been made to replicate or extend natural hazard studies concerned with the various factors that mediate between perceived risk and actual behavior, including the perceived benefits of risk mitigation actions, the presence or absence of evidence validating or confirming the threat, and the individual's mental image of potential damages.

Finally, the findings reported in this paper are confounded by several unresolved problems: risk perceptions may change rapidly; people may not understand how their perceptions and preferences translate into policy; and people may prefer alternatives not realistically obtainable. With these reservations and qualifications in mind, some of the major findings of risk perception literature are discussed below.

### **Human Intellectual Limitations**

Research suggests that people do not cope well when confronted with risk problems and decisions. Intellectual limitations and the need to reduce anxiety often lead to the denial that risk and uncertainty exist and to unrealistic oversimplifications of essentially complex problems [120, 127, 150]. To simplify risk problems, people use a number of inferential or judgmental rules, known technically as heuristics [54, 55, 147, 148, 149, 150]. Two of the most important are 1) information availability, or the tendency for people to judge an event more frequent if instances of it are easy to imagine or recall; and 2) representativeness, or the tendency of people to assume that roughly similar activities and events (such as nuclear power technologies and nuclear war) have the same characteristics and risks. These judgmental operations enable people to reduce difficult probabilistic and assessment tasks to simpler tasks; however, these judgmental operations also lead to severe and systematic biases and errors.

One bias associated with information availability is that people have difficulty imagining low probability/high consequence events happening to themselves [9, 120, 127]. Unless people have been made graphically aware of the risks, typically through past experience, they are unlikely to take protective action [63, 64, 165]. A classic example is the observed reluctance of floodplain residents to purchase low-cost flood insurance [64]. Compounding the problem is the difficulty people have understanding and interpreting probabilistic information. People residing in 100-year floodplains, for example, typically believe that a recent severe flood precludes the possibility of another severe flood in the near future [9, 165]. According to folk wisdom but not to probability theory, lightning never strikes the same place twice.

Biases associated with information availability have also been used to explain the results of studies in which people were asked to judge the frequency of various causes of death, such as accidents, tornadoes, and diseases [127]. These studies show that the risks of low-frequency events tend to be overestimated, and that the risks of high-frequency events are underestimated. People underestimate fatalities caused by asthma, stroke, and diabetes, and overestimate fatalities from homicides, botulism, fires, snake bites, tornadoes, and abortion. Overestimated causes of death tend to be dramatic, sensational, and of interest to the media, whereas underestimated causes of death tend to be unspectacular, undramatic, and of little interest to the media [14, 127].

Researchers have also pointed out that information availability biases may cause risk information campaigns and educational efforts to work at cross-purposes [133]. Information may heighten the imaginability and consequently the perceived probability of a rare

event, even when the information is designed to assure individuals that the event is unlikely. A package insert listing all the risks of taking a drug, or a published report describing safety precautions taken at a DNA research laboratory, may serve only to increase concern about the substance or activity. By identifying previously unknown ways in which things can go wrong, the information provider takes the chance that people will incorrectly assess the information (i.e., that they will consider the event more likely as a result of increased knowledge). As one observer notes [77, p. 61]:

We generally assume that informed advice is valuable to political policy-makers. However, in the context of a controversial political issue, and when the relevant technical analysis is ambiguous, then the value of scientific advice become questionable. A technical controversy sometimes creates confusion rather than clarity, and it is possible that the dispute itself may become so divisive and widespread that scientific advice becomes more of a cost than a benefit to the policy-maker and society.

Unfortunately, few researchers have critically examined the controversial hypothesis implicit in this work—that the very discussion of a low-probability hazard increases the judged probability of the hazard, regardless of what the evidence indicates.

Disputes and controversies about risk are made all the more difficult by another psychological mechanism: once beliefs are formed, individuals frequently structure and distort the interpretation of new evidence and often resist disconfirming information [116, 127]. People tend to dismiss evidence contradicting their beliefs as unreliable, erroneous, and unrepresentative. The accident at Three Mile Island, for example, provided confirming evidence for those already convinced that nuclear power technology is safe [90]; the accident also reinforced the beliefs of those who believed that nuclear power technology is dangerous. Convincing people that a hazard they fear is not a hazard is extremely difficult even under the best conditions. Any accident or mishap, no matter how small, is seen as proof of high risk [133].

### **Overconfidence**

Another second set of risk perception findings addresses the problem of overconfidence. Researchers have shown that experts and laypersons are typically overconfident about their risk estimates. In one study participants were asked to state the odds that they were correct in judging which of two lethal events was the more frequent [127]. Most people claimed that the odds of their being wrong were 100:1 or greater. In actuality, people were wrong about one out of every eight times. Such overconfidence can produce serious judgmental errors, including judgments about how much is known about the hazard and about how much needs to be known. Of equal or greater importance, overconfidence leads people to believe that they are comparatively immune to common hazards. Studies show that 1) most people rate themselves among the most skillful and safe drivers in the population; 2) people rate their own personal risk from several common household hazards as lower than the risk for others in society; 3) people judge themselves average or above average in their ability to avoid bicycle and power lawnmower accidents; and 4) people underestimate and are extremely unrealistic about their chances of having a heart attack [115, 139, 158]. In general, people underestimate the risks of activities that they perceive to be familiar and under their personal control, such as automobile driving.

Overconfidence has also been used to explain in part the observed reluctance of about 80 to 90% of the U.S. driving population to wear seat belts [124]. Unfortunately, few empirical studies have examined this issue or the more general relationship between perceived risk and protective behavior. One new study by Slovic, Lichtenstein, and

Fischhoff of seat belt usage is, however, exploring and empirically testing the hypothesis that seat belt usage will increase if the public is presented with information about the lifetime risks of driving instead of information about the risks of taking a single trip.

### **Expert and Nonexpert Estimates of Risk**

A third set of findings bears on expert and nonexpert estimates of risk. A consistent result is that technical experts and nonexperts differ substantially in their risk estimates [57]. Risk estimates of technical experts are closely correlated with annual fatality rates, whereas the risk estimates of nonexperts are only moderately to poorly correlated with annual fatality rates [129]. In explaining these differences, researchers have identified several factors other than annual fatality rates that influence public perceptions of risk [72, 127, 153, 154]. Risks are perceived to be higher if the activity is perceived to be involuntary, catastrophic, not personally controllable, inequitable in the distribution of its risk and benefits, unfamiliar, and highly complex. Other factors influencing risk perceptions are whether the adverse effects are immediate or delayed, whether exposure to the hazard is continuous or occasional, whether the technology is perceived to be necessary or luxury, whether the adverse effects are well-known or uncertain, and whether the activity is certain to be fatal.

Several studies have shown that these dimensions of risk are closely related to each other [126, 153, 154]. Such correlations have prompted several research groups to reduce the various dimensions of risk to a smaller number of factors. One study identified at least two factors [29]: the level of technological complexity and the hazard's severity or catastrophic potential. In a follow-up study that examined a larger set of hazards and risk characteristics, Slovic, Fischhoff, and Lichtenstein [127] found three factors: familiarity, dread, and the number of people exposed to the hazard. In an ongoing European study of risk perception, Vlek and Stallen identified several additional factors influencing risk perception and risk acceptability, including the beneficiality of the technology and the degree to which protection is provided by institutional means [153, 154]. In spite of these different findings, it is clear that a hazard's catastrophic potential is uppermost in the minds of people. Because catastrophic events may threaten the survival of individuals, families, societies, and the species as a whole, such concern may be quite justifiable.

Analysis of intercorrelations between the various dimensions of risk have also led researchers to challenge Starr's [135] well-known proposition that the risks of voluntary activities are more acceptable to the general public than the risks of involuntary activities [5, 99, 127]. One problem with this proposition is that voluntary risks are also perceived by the public to be controllable, equitable, familiar, and noncatastrophic. These correlations suggest in turn that the observed greater willingness of the public to accept voluntary risks may be due to these other factors and not to the voluntary nature of the activity.

It appears that differences between expert and nonexpert perceptions of risk may be at least partially rooted in the different risk analysis methods and approaches used to assess and evaluate risks [1]. Technical experts often implicitly and sometimes explicitly assign equal weight to hazards that take many lives at one time and to hazards that take many lives one at a time; nonexperts typically assign greater weight to hazards that take many lives at one time (e.g., catastrophes). Technical experts often implicitly and sometimes explicitly assign equal weight to statistical and known deaths; nonexperts typically assign greater weight to known deaths. It is interesting in this regard to note the high levels of public concern and massive allocations of resources devoted to rescuing an identifiable

person lost at sea. Technical experts often implicitly and sometimes explicitly assign equal weight to voluntary and involuntary risks; nonexperts typically assign greater weight to involuntary risks. Technical experts typically express risks in quantitative terms and use computational and experimental methods to identify, estimate, and evaluate the risks; nonexperts typically express risks in qualitative terms and use intuitive and impressionistic methods to identify, estimate, and evaluate the risks. Technical experts typically believe that quantitative estimates of risk should be the prime consideration in risk acceptability decisions; nonexperts typically believe that quantitative estimates of risk should be only one among several quantitative and qualitative considerations in risk acceptability decisions. Technical experts often implicitly and sometimes explicitly assign the same weight to different ways of dying; nonexperts typically feel that some ways of dying are worse than others. How one dies, and with how much suffering, is as important as where and when.

### **Risk Perception and Nuclear Power: A Case Study**

To date, most of the research on risk perception has focused on nuclear power [7, 18, 32, 35, 50, 58, 76, 80, 87, 88, 90, 92, 93, 94, 101, 103, 105, 106, 108, 133, 134, 142, 143, 156, 157]. These studies have produced several important findings. First, researchers have shown that nuclear power has nearly all the characteristics associated with high perceived risk. The risks of nuclear power are perceived to be involuntary, delayed in their consequences, unknown, uncontrollable, unfamiliar, potentially catastrophic, inequitable, and certain to be fatal [133]. Public perceptions of nuclear power contrast sharply with nonnuclear sources of electric power, which are perceived to be noncatastrophic, familiar, controllable, and comparatively safe.

Second, researchers have shown that disputes about nuclear power are often about values and goals that far transcend issues of health and safety [80, 90, 92, 100, 101, 134]. Many people are concerned about nuclear power not because of its specific risks but because of its associations with nuclear weapons, highly centralized political and economic systems, and technological elitism. The debate about nuclear power is also colored by social class—people with lower socioeconomic status are less supportive of nuclear power than those with higher socioeconomic status; by sex—women are less supportive of nuclear power than men are; and by concerns about the credibility of institutions charged with estimating, evaluating, and managing the risks [80].

Despite these concerns, research studies consistently show that the public, by a margin of 2 and sometimes 3 to 1, supports nuclear power, even in the aftermath of Three Mile Island [49, 80]. Somewhat counterintuitively, researchers have also found that people living within the vicinity of a nuclear power plant (and therefore presumably subject to the greatest objective risk) are more supportive of nuclear power than those living farther away [80]. In explaining this finding it has been proposed that people living near power plants receive greater economic benefits, that they experience greater cognitive dissonance, and that they have had their worst fears assuaged by a history of accident-free operations. Interestingly, those who are least supportive of nuclear power live in areas where power plants are under construction or being planned. One policy implication arises from these findings. In several countries, including France, proposals are currently being considered to compensate those who live in the vicinity of nuclear power plants. If the intention is to win wider public acceptance, then the policy is misdirected. Those living nearest to the power plant are already supportive and little would be gained by compensat-

ing them. By comparison, compensating those who are least supportive (i.e., those living in areas where power plants are under construction or being planned) might have a major impact. Such a policy of course, could also backfire by providing support for the belief that the risks of nuclear power are indeed substantial.

What this specific case and others similar to it teach us is that analysts and decision-makers need a better understanding of how people think and make decisions about technological risks. Public risk acceptance and the success of risk management policies are likely to hinge on such understanding. Stated more forcefully, without such understanding well-intended policies may be ineffective or even counterproductive.

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